

Education

University of California, Santa Cruz

Ph.D. | Computer Science | Adv. Dr. **N. Norouzi**, UC Berkeley | 3.92 GPA

Santa Cruz | Berkeley, CA

Grad. 2026

University of Puerto Rico, Rio Piedras

B.Sc. | Computer Science | PI. Dr. **R. Mégret** | 3.75 GPA

San Juan, PR

Grad. 2021

University of Naples, Federico II

Apple Developer Academy, Software Engineering & Design

Naples, Italy

Grad. 2019

Experience

Tera AI

Founding Lead Research Scientist | Generative AI, Spatial Reasoning & Autonomy

San Francisco, CA

June 2024 – Present

- Designed and executed research agenda, aligning ambitious experiments with fast iteration timelines, budgets and compute. Research highlight: single transformer architecture capable of novel multimodal geospatial autonomy

Google Research

Student Researcher (PhD) | Computer Vision, Large Language Models

Mountain View, CA

June 2023 – September 2023

- Established foundation models for unified multi-modal video & image training, unlocking significantly higher performance ($\uparrow >25\%$) compared to previously deployed single-modality encoders

Google X

AI Resident (PhD) | Simulation, Edge Perception, Aerial Imaging

Mountain View, CA

June 2021 – September 2022

- Exceeded perf. expectations for aerial & ground perception systems under novel problem domains, granted patents

University of California, Santa Cruz

Graduate Student Researcher | Medical AI Research, Generative AI

Santa Cruz, CA

June 2020 – June 2026

- Achieved SOTA performance ($\uparrow 18\%$) via generative methods (e.g. Diffusion) for image segmentation, classification and analysis in label-constrained scenarios (< 100 samples); published multiple first-author papers

Publications & Conference Presentations

- H. Carrión**, **N. Norouzi**. DermDepth: Toward Monocular Metric Scale 3D Reconstruction Models for Dermatology, MICCAI 2026
- H. Carrión**, **N. Norouzi**. cgDDI: Controllable Generation of Diverse Dermatological Imagery for Fair and Efficient Malignancy Classification, MICCAI 2026
- H. Carrión**, **P. Perona**, **J. Malik**, et al. TARDIS STRIDE: A Spatio-Temporal Road Image Dataset and World Model for Autonomy, Preprint, 2025
- H. Carrión**, **N. Norouzi**. FEDD - Fair, Efficient, and Diverse Diffusion-based Lesion Segmentation and Malignancy Classification, MICCAI 2023
- H. Carrión**, et al. Patch HealNet: Predicting Wound Stage from In Vivo Imaging, DARPA BETR Review 2023
- H. Carrión**, et al. HealNet - Self-Supervised Acute Wound Heal-Stage Classification, MICCAI MLMI 2022
- H. Carrión**, et al. Automatic Wound Detection and Size Estimation using Deep Learning Algorithms, PLOS Computational Biology 2022
- J. Chan, **H. Carrión**, **R. Mégret** et al. Honeybee Re-identification in Video: New Datasets and Impact of Self-supervision, VISAPP 2022
- H. Carrión**, et al. Towards Classification of Wound Stages Using Deep Learning Algorithms, SACNAS 2020

Skills, Awards & Advisorships

Technical: Python | C++ | Swift | PyTorch | TensorFlow | Keras | JAX | Numpy | Natural Language Processing | SW Eng

Soft: Strong communicator, presenter, collaborator, diligent, organized **Languages:** English (native) | Spanish (native)

Awards: Google-CAHSI Fellowship | MICCAI STAR | SIDIM Outstanding Research | NIH BD2K | Google Patent Award

Advisorships: Technical Advisor @ visia.ai (2021-Present) | ML Tutor (2020-2022) **Ethics:** Real-world positive impact